

Problems in Regression

1)

Obtain the equations of the lines of regression from the following data:

X:	1	2	3	4	5	6	7
Y:	9	8	10	12	11	13	14

Ans:

$$3.7x - 4y + 29.2 = 0 \quad 4x - 3.7y + 24.7 = 0$$

2)

Obtain the equations of the regression lines from the following data, using the method of least squares. Hence find the coefficient of correlation between X and Y. Also estimate the value of (i) Y, when $X = 38$ and (ii) X, when $Y = 18$.

Solution	X:	22	26	29	30	31	31	34	35
	Y:	20	20	21	29	27	24	27	31

Answer:

$$y = 0.83x + 0.18$$

$$x = 0.81y + 9.60$$

31.72 24.18

3)

A study of prices of rice at Chennai and Madurai gave the following data:

	Chennai	Madurai
Mean	19.5	17.75
S.D.	1.75	2.5

Also the coefficient of correlation between the two is 0.8. Estimate the most likely price of rice (i) at Chennai corresponding to the price of 18 at Madurai and (ii) at Madurai corresponding to the price of 17 at Chennai.

Ans: 19.64, 14.89

4)

In a partially destroyed laboratory record of an analysis of correlation data, the following results only are legible: Variance of $X = 1$. The regression equations are $3x + 2y = 26$ and $6x + y = 31$. What were (i) the mean values of X and Y? (ii) the standard deviation of Y? and (iii) the correlation coefficient between X and Y?

5)

Given that $x = 4y + 5$ and $y = kx + 4$ are the regression lines of X on Y and of Y on X respectively, show that $0 \leq k \leq \frac{1}{4}$. If $k = \frac{1}{16}$, find the means of X and Y and r_{XY} .

6)

Find the angle between the two lines of regression. Deduce the condition for the two lines to be (i) at right angles and (ii) coincident.

7)

Find the equations of the regression lines from the following data. Hence calculate the coefficient of correlation between X and Y .

X :	62	64	65	69	70	71	72	74
Y :	126	125	139	145	165	152	180	208

8)

Find the equations of the regression lines from the following data. Also estimate the value of Y when $X = 71$ and the value of X when $Y = 70$.

X :	65	66	67	67	68	69	70	72
Y :	67	68	65	68	72	72	69	71

9)

Find the equation of the regression line of Y on X using the method of least squares from the following data. Find the value of Y corresponding to $X = 18$.

X :	5	10	15	20	25
Y :	16	19	23	26	30

10)

Obtain the line of regression of X on Y using the method of least squares from the following data. Find the value of X when $Y = 45$.

X :	4.7	8.2	12.4	15.8	20.7	24.9	31.9	35.0	39.1	38.8
Y :	4.0	8.0	12.5	16.0	20.0	25.0	31.0	36.0	40.0	40.0